

ENTERPRISE ARCHITECTURE

THE ZACHMAN FRAMEWORK: SOLVING GENERAL MANAGEMENT PROBLEMS

JOHN A. ZACHMAN
ZACHMAN INTERNATIONAL

AGENDA

- I. The Zachman Framework
- II. A Zachman Framework Story
- III. An Illustration of Primitive Modeling
Concepts
- IV. Methodology for Solving General
Management Problems
- V. Conclusions

THE INFORMATION AGE

"The next information revolution is well underway. But it is not happening where information scientists, information executives, and the information industry in general are looking for it. It is not a revolution in technology, machinery, techniques, software, or speed. It is a revolution in CONCEPTS."

Peter Drucker. Forbes ASAP, August 24, 1998

INTRODUCTION TO
ENTERPRISE ARCHITECTURE

THE FRAMEWORK FOR ENTERPRISE ARCHITECTURE

JOHN A. ZACHMAN
ZACHMAN INTERNATIONAL

The Zachman Framework for Enterprise Architecture™

The Enterprise Ontology™

Version 3.0



FRAMEWORK GRAPHIC

For the latest version of the Framework Graphic,
register at [**www.Zachman.com**](http://www.Zachman.com)
for a high resolution .pdf file.

(For a publication release of the Framework Graphic
send requests to the Contact Us link on zachman.com)

You may be interested in several articles by John A. Zachman at
Zachman.com

“Architecture Is Architecture Is Architecture”

“John Zachman’s Concise Definition of the Zachman Framework”
and

“The Zachman Framework Evolution” by John P. Zachman

ENGINEERING VS MANUFACTURING

Engineering work requires

single-variable,

(**Synthesis**)

ontologically-defined descriptions

of the *whole* of the object.

(Primitive)

(**This is RADICALLY different**)

IN CONTRAST

Manufacturing work requires

multi-variable,

holistic descriptions

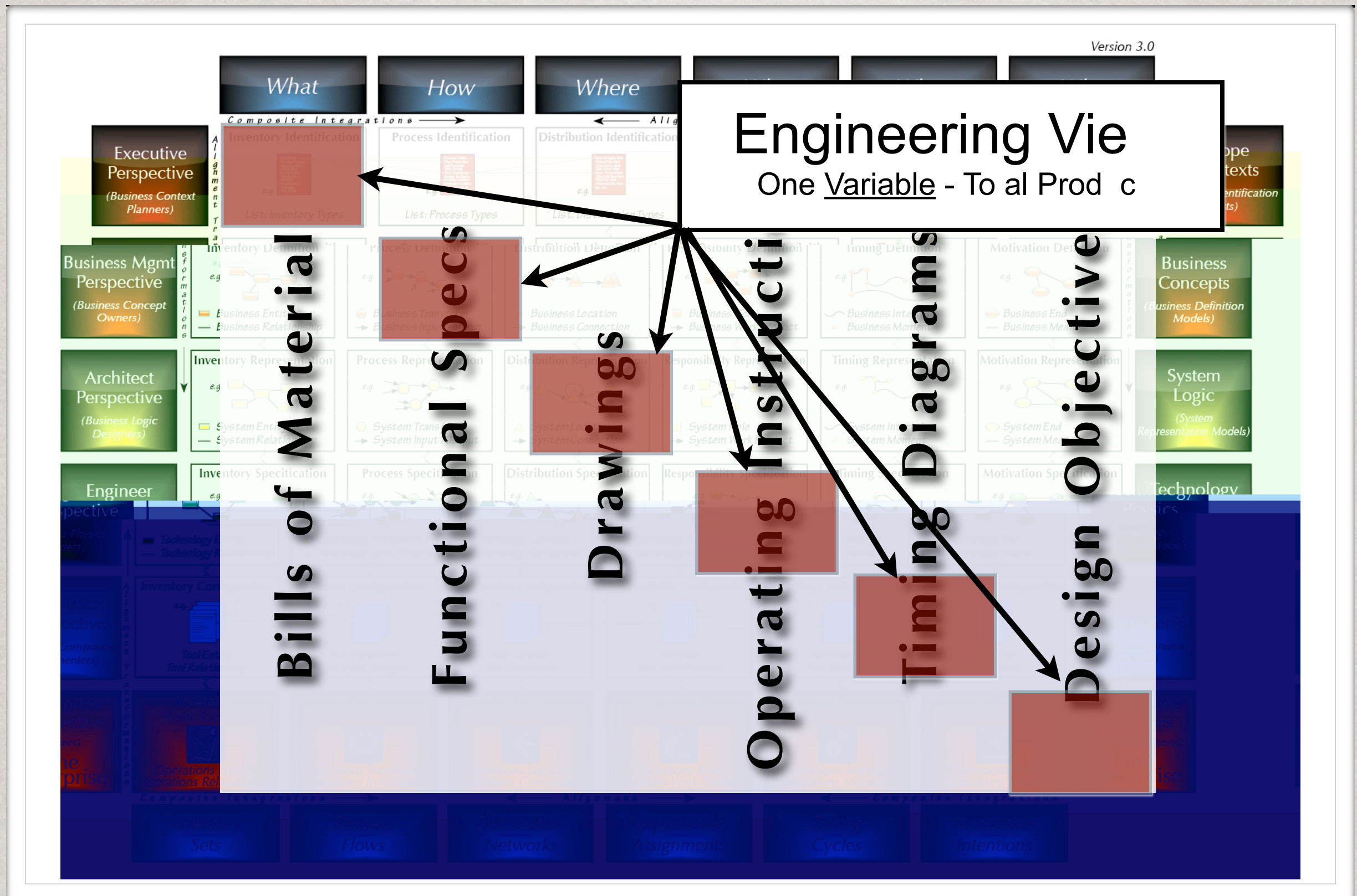
of *parts* of the object.

(**Analysis**)

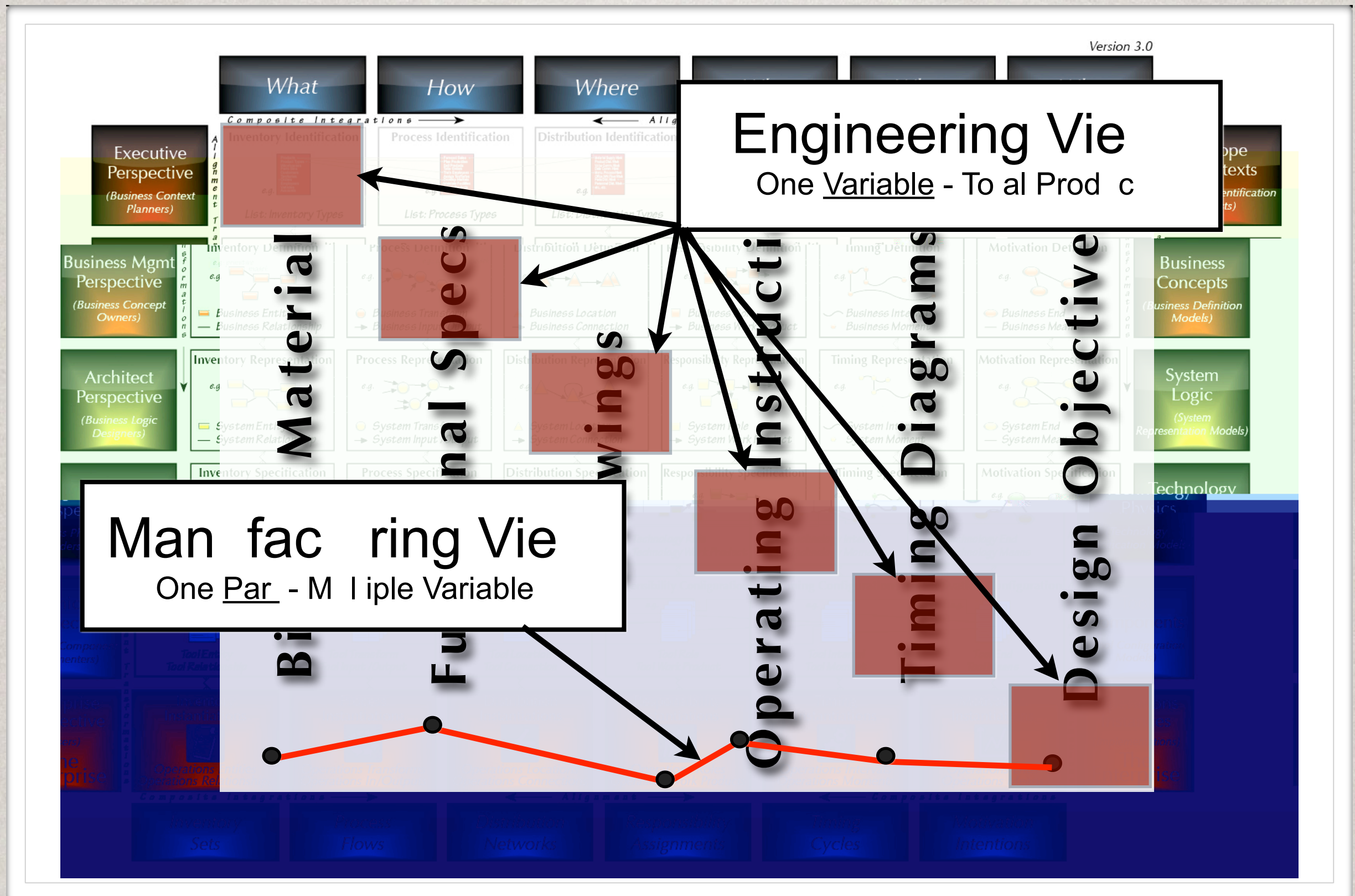
(Composite)

(**This is STANDARD practice**)

ENGINEERING VERSUS MANUFACTURING



ENGINEERING VERSUS MANUFACTURING



ONTOLOGY

The Zachman Framework™ schema technically is an ontology -
a theory of the existence of a structured set
of essential components of an object
for which explicit expression is necessary (is mandatory?)
for designing, operating and changing the object
(the object being an Enterprise, a department, a value chain,
a "sliver," a solution, a project,
an airplane, a building, a bathtub or whatever or whatever).

A Framework is a STRUCTURE.
(A Structure DEFINES something.)

METHODOLOGY

A Methodology is a PROCESS.
(A Process TRANSFORMS something.)

A Structure IS NOT A Process
A Process IS NOT a Structure.

ONTOLOGY VS METHODOLOGY

An Ontology is the classification of the total set of “**Primitive**” (elemental) components that exist and that are relevant to the existence of an object.

A Methodology produces “**Composite**” (compound) implementations of the Primitives.

Primitives (elements) are timeless.

Composites (compounds) are temporal.

ONTOLOGY

PERIODIC TABLE OF THE ELEMENTS

<http://www.ktf-split.hr/periodni/en/>

Legend:

- Metal
- Semimetal
- Nonmetal
- 1 Alkali metal
- 2 Alkaline earth metal
- 3-10 Transition metals
- Lanthanide
- Actinide
- 16 Chalcogens element
- 17 Halogens element
- 18 Noble gas

Standard State (25 °C, 101 kPa):

- Ne - gas
- Fe - solid
- Ga - liquid
- Tc - synthetic

Periodic Table Data (Selected Elements):

Period	Group	Element	Symbol	Atomic Number	Relative Atomic Mass (1)
1	IA	Hydrogen	H	1	1.0079
1	VIIIA	Helium	He	2	4.0026
2	IIA	Beryllium	Be	4	9.0122
2	IIIA	Boron	B	5	10.811
2	IVB	Carbon	C	6	12.011
2	V	Nitrogen	N	7	14.007
2	VI	Oxygen	O	8	15.999
2	VII	Fluorine	F	9	18.998
2	VIII	Neon	Ne	10	20.180
3	IA	Lithium	Li	3	6.941
3	IIA	Beryllium	Be	4	9.0122
3	IIIA	Boron	B	5	10.811
3	IVB	Carbon	C	6	12.011
3	V	Nitrogen	N	7	14.007
3	VI	Oxygen	O	8	15.999
3	VII	Fluorine	F	9	18.998
3	VIII	Neon	Ne	10	20.180
3	IX	Sodium	Na	11	22.990
3	X	Magnesium	Mg	12	24.305
3	XI	Aluminum	Al	13	26.982
3	XII	Silicon	Si	14	28.086
3	XIII	Phosphorus	P	15	30.974
3	XIV	Sulfur	S	16	32.065
3	XV	Chlorine	Cl	17	35.453
3	XVI	Argon	Ar	18	39.948
4	IA	Potassium	K	19	39.098
4	IIA	Calcium	Ca	20	40.078
4	IIIB	Scandium	Sc	21	44.956
4	IVB	Titanium	Ti	22	47.867
4	V	Vanadium	V	23	50.942
4	VI	Chromium	Cr	24	51.996
4	VII	Manganese	Mn	25	54.938
4	VIII	Iron	Fe	26	55.845
4	IX	Cobalt	Co	27	58.933
4	X	Nickel	Ni	28	58.693
4	XI	Copper	Cu	29	63.546
4	XII	Zinc	Zn	30	65.39
4	XIII	Gallium	Ga	31	69.723
4	XIV	Germanium	Ge	32	72.64
4	XV	Arsenic	As	33	74.922
4	XVI	Selenium	Se	34	78.96
4	XVII	Bromine	Br	35	79.904
4	XVIII	Krypton	Kr	36	83.80
5	IA	Rubidium	Rb	37	85.468
5	IIA	Strontium	Sr	38	87.62
5	IIIB	Yttrium	Y	39	88.906
5	IVB	Zirconium	Zr	40	91.224
5	V	Niobium	Nb	41	92.906
5	VI	Molybdenum	Mo	42	95.94
5	VII	Technetium	Tc	43	(98)
5	VIII	Ruthenium	Ru	44	101.07
5	IX	Rhodium	Rh	45	102.91
5	X	Palladium	Pd	46	106.42
5	XI	Silver	Ag	47	107.87
5	XII	Cadmium	Cd	48	112.41
5	XIII	Indium	In	49	114.82
5	XIV	Tin	Sn	50	118.71
5	XV	Antimony	Sb	51	121.76
5	XVI	Tellurium	Te	52	127.6
5	XVII	Iodine	I	53	126.905
5	XVIII	Xenon	Xe	54	131.29
6	IA	Cesium	Cs	55	132.91
6	IIA	Barium	Ba	56	137.33
6	IIIB	Lanthanum	La	57	138.91
6	IVB	Hafnium	Hf	72	178.49
6	V	Tantalum	Ta	73	180.95
6	VI	Tungsten	W	74	183.84
6	VII	Rhenium	Re	75	186.21
6	VIII	Osmium	Os	76	190.23
6	IX	Iridium	Ir	77	192.22
6	X	Platinum	Pt	78	195.08
6	XI	Gold	Au	79	196.97
6	XII	Mercury	Hg	80	200.59
6	XIII	Thallium	Tl	81	204.38
6	XIV	Lead	Pb	82	207.2
6	XV	Bismuth	Bi	83	208.98
6	XVI	Polonium	Po	84	(209)
6	XVII	Astatine	At	85	(210)
6	XVIII	Radon	Rn	86	(222)
7	IA	Francium	Fr	87	(223)
7	IIA	Radium	Ra	88	(226)
7	IIIB	Actinium	Ac	89	(227)
7	IVB	Rutherfordium	Rf	104	(261)
7	V	Dubnium	Db	105	(262)
7	VI	Seaborgium	Sg	106	(266)
7	VII	Bohrium	Bh	107	(264)
7	VIII	Hassium	Hs	108	(277)
7	IX	Meitnerium	Mt	109	(268)
7	X	Ununnilium	Uun	110	(289)
7	XI	Ununnilium	Uun	111	(288)
7	XII	Ununnilium	Uun	112	(295)
7	XIII	Ununnilium	Uun	113	(284)
7	XIV	Ununnilium	Uun	114	(289)
7	XV	Ununnilium	Uun	115	(288)
7	XVI	Ununnilium	Uun	116	(289)
7	XVII	Ununnilium	Uun	117	(284)
7	XVIII	Ununnilium	Uun	118	(289)

Lanthanide Series:

Element	Symbol	Atomic Number	Relative Atomic Mass (1)
Lanthanum	La	57	138.91
Cerium	Ce	58	140.12
Praseodymium	Pr	59	140.91
Neodymium	Nd	60	144.24
Promethium	Pm	61	(145)
Samarium	Sm	62	150.36
Europium	Eu	63	151.96
Gadolinium	Gd	64	157.25
Terbium	Tb	65	158.93
Dysprosium	Dy	66	162.50
Holmium	Ho	67	164.93
Erbium	Er	68	167.26
Thulium	Tm	69	168.93
Ytterbium	Yb	70	173.04
Lutetium	Lu	71	174.97

Actinide Series:

Element	Symbol	Atomic Number	Relative Atomic Mass (1)
Actinium	Ac	89	(227)
Thorium	Th	90	232.04
Protactinium	Pa	91	231.04
Uranium	U	92	238.03
Neptunium	Np	93	(237)
Plutonium	Pu	94	(244)
Americium	Am	95	(243)
Curium	Cm	96	(247)
Berkelium	Bk	97	(247)
Californium	Cf	98	(251)
Einsteinium	Es	99	(252)
Fermium	Fm	100	(257)
Mendelevium	Md	101	(258)
Nobelium	No	102	(259)
Lawrencium	Lr	103	(262)

Footnote (1): Pure Appl. Chem., 73, No. 4, 667-683 (2001). Relative atomic mass is shown with five significant figures. For elements having no stable nuclei, the value enclosed in brackets indicates the mass number of the longest-lived isotope of the element. However three such elements (Th, Pa, and U) do have a characteristic terrestrial isotopic composition, and for these an atomic weight is tabulated.

Editor: Aditya Vardhan (adivardhan@rediffmail.com)

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This is NOT a Process.

Elements are Timeless

Until an ontology exists, nothing is repeatable, nothing is predictable.
There is no DISCIPLINE.

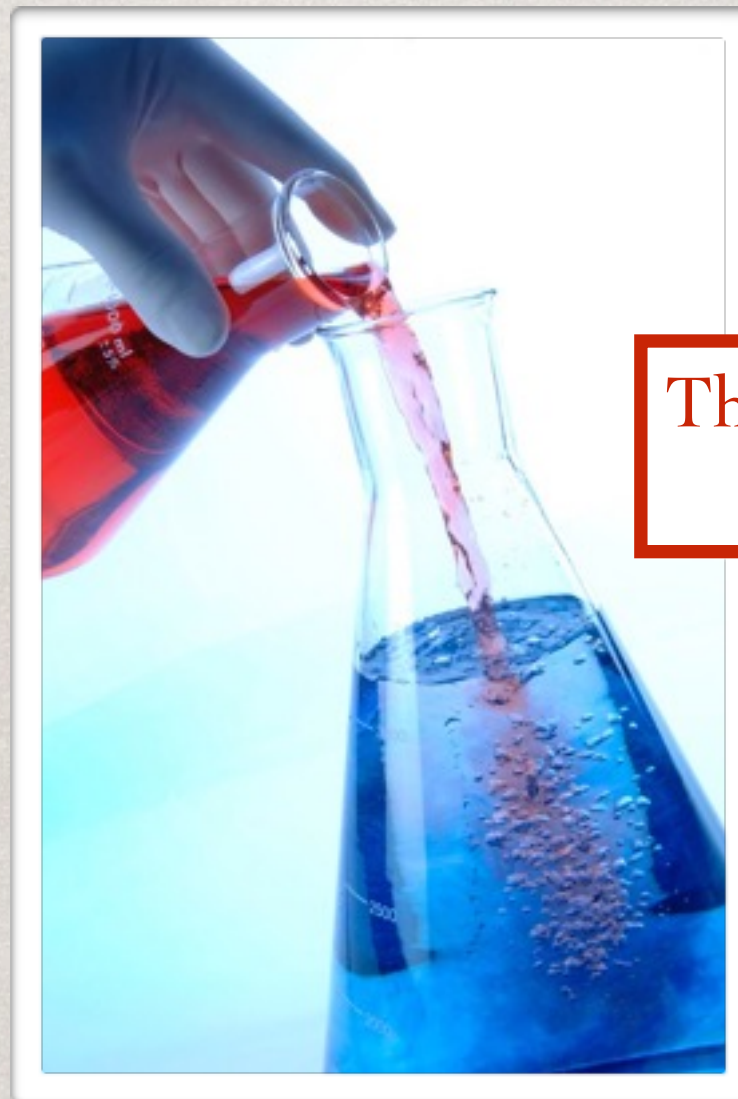
PROCESS

(Methodology)

A Process TRANSFORMS something.

This is a Process:

Add Bleach to
an Alkali and
it is
transformed
into Saltwater.



This is NOT an
Ontology.

Compound are Temporal

PROCESS

(METHODOLOGY)

Add Bleach to an Alkali and
it is transformed into Saltwater.

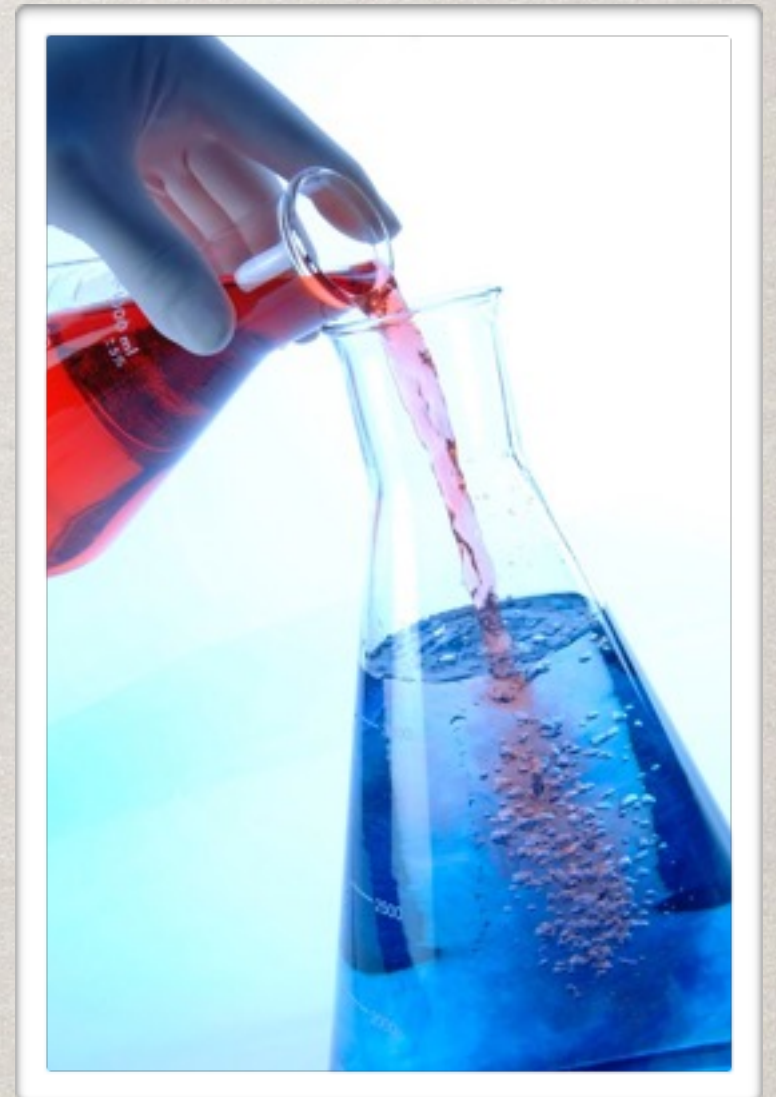


COMPOUNDS

Salt	NaCl
Aspirin	$\text{C}_9\text{H}_8\text{O}_4$
Vicodin	$\text{C}_{18}\text{H}_{21}\text{NO}_3$
Naproxen	$\text{C}_{14}\text{H}_{14}\text{O}_3$
Ibuprophen	$\text{C}_{13}\text{H}_{18}\text{O}_2$
Viagra	$\text{C}_{22}\text{H}_{30}\text{N}_6\text{O}_4\text{S}$
Sulphuric Acid	H_2SO_4
Water	H_2O

etc., etc., etc.

Compound are Temporal



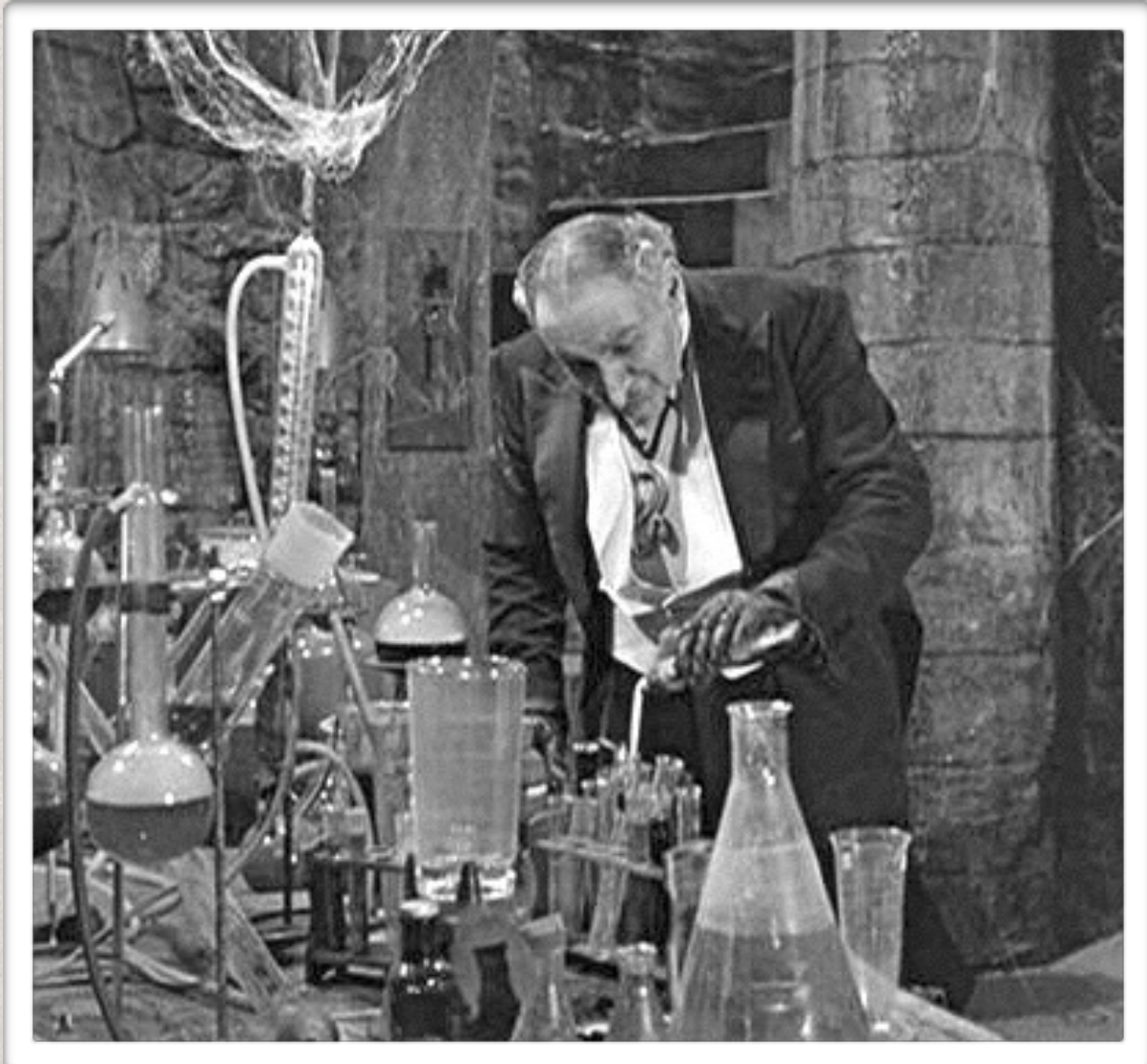
ALCHEMY - A PRACTICE

This is a Methodology WITHOUT an Ontology

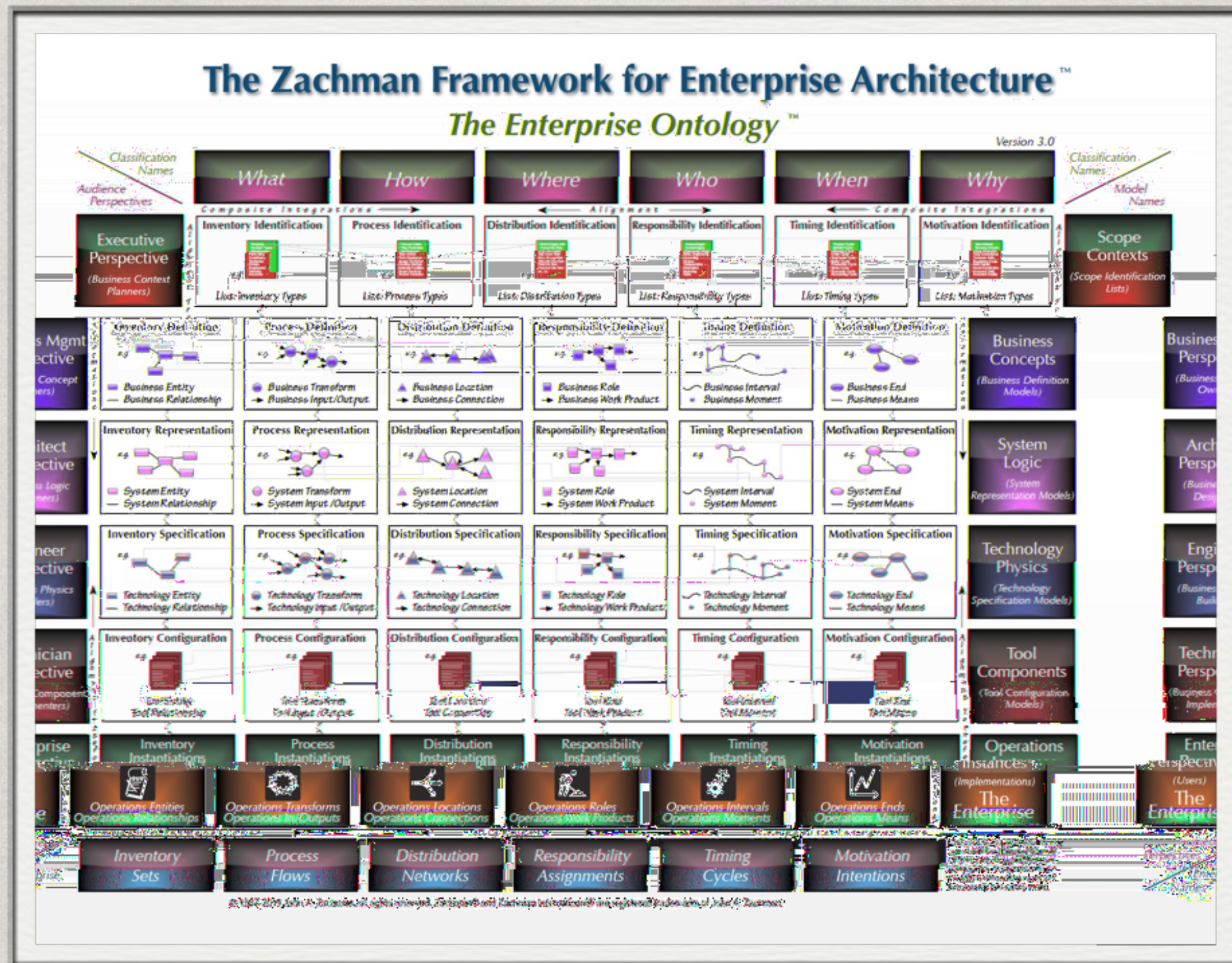
A Process with no ontological structure is ad hoc, fixed and dependent on practitioner skills.

This is NOT a science.

It is ALCHEMY, a "**practice**."



ONTOLOGY



Primi i e are Timele .

Until an ontology exists, nothing is repeatable, nothing is predictable.
There is no DISCIPLINE.

PROCESS

(METHODOLOGY)

COMPOSITES

(COMPOUNDS)

COBOL Programs

Objects

BPMN Models

Swimlanes

Business Architecture

Capabilities

Mobility

Applications

Data Models

Security Architecture

Services

COTS

Technology Architecture

Big Data

Missions/Visions

Agile Code

Business Processes

DoDAF Models

Balanced Scorecard

Clouds

I.B. Watson

TOGAF Artifacts

Etc., etc., etc.

Compound are Temporal

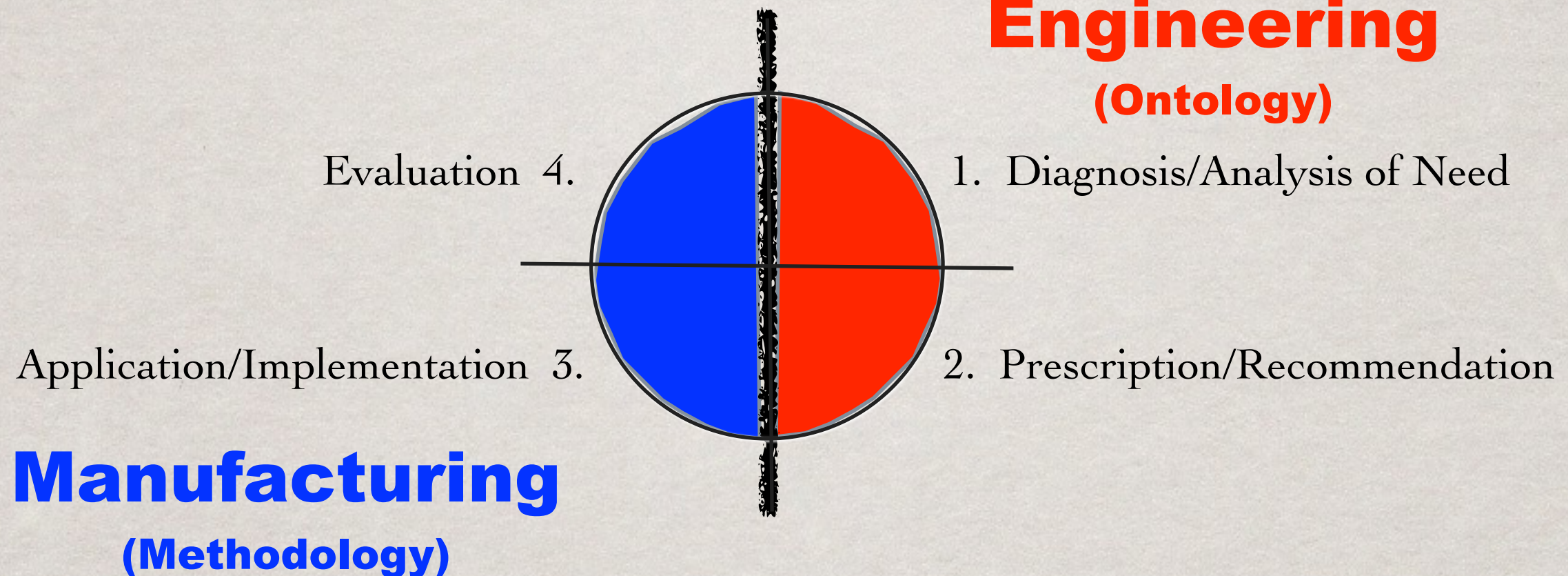
INTRODUCTION TO
ENTERPRISE ARCHITECTURE

A
ZACHMAN FRAMEWORK
STORY

JOHN A. ZACHMAN
ZACHMAN INTERNATIONAL

Professional Service Cycle

Engineering
(Ontology)



Roger Greer:

Dean

School of Library and Information Management

University of Southern California

(My notes from a 1991, IBM GUIDE Conference presentation)

INTRODUCTION TO
ENTERPRISE ARCHITECTURE

ILLUSTRATIONS
OF
PRIMITIVE MODELING
CONCEPTS

JOHN A. ZACHMAN
ZACHMAN INTERNATIONAL

Executive
Personnel
(Business
Plan)



Business
Personnel
(Business
Concept)



Account
Personnel
(Business
Development)



Finance
Personnel
(Business
Analysis)

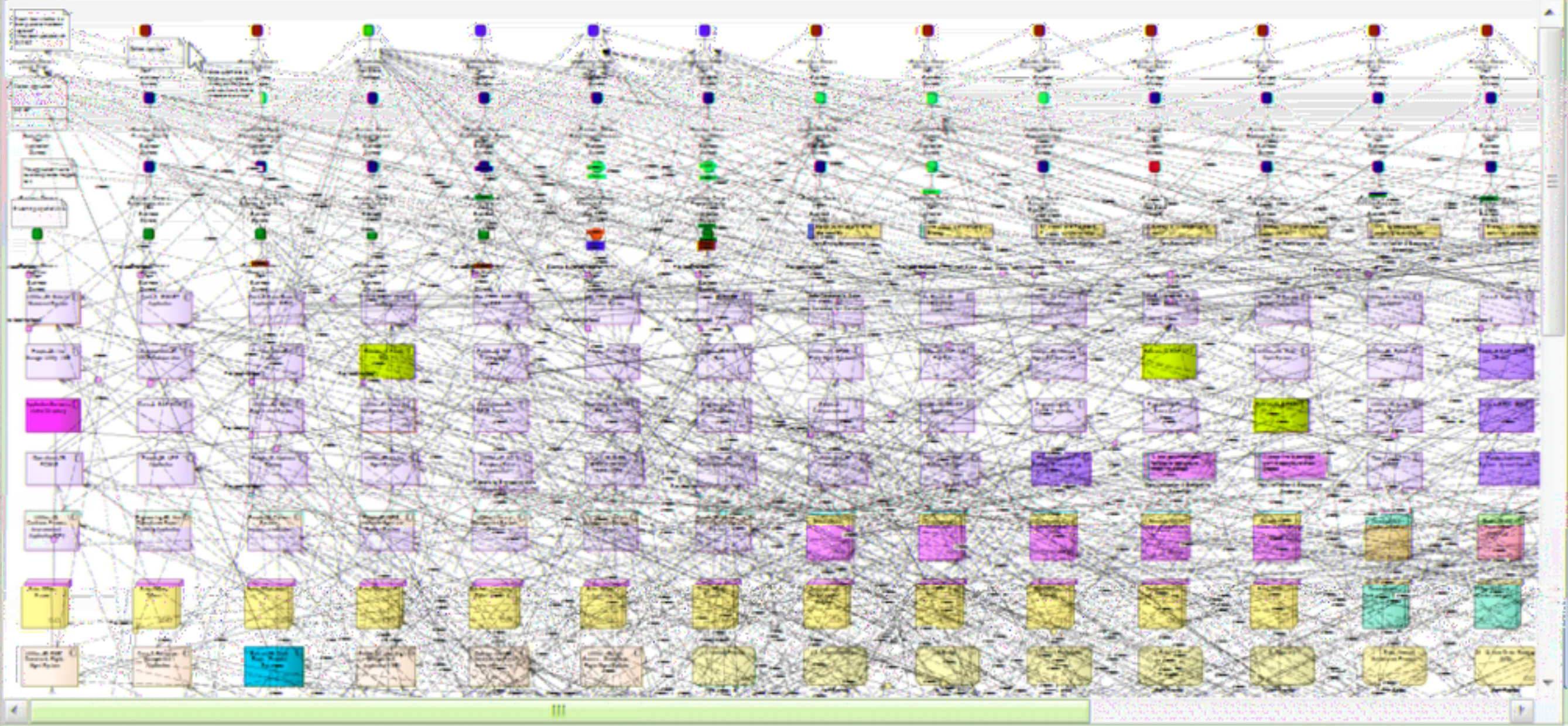


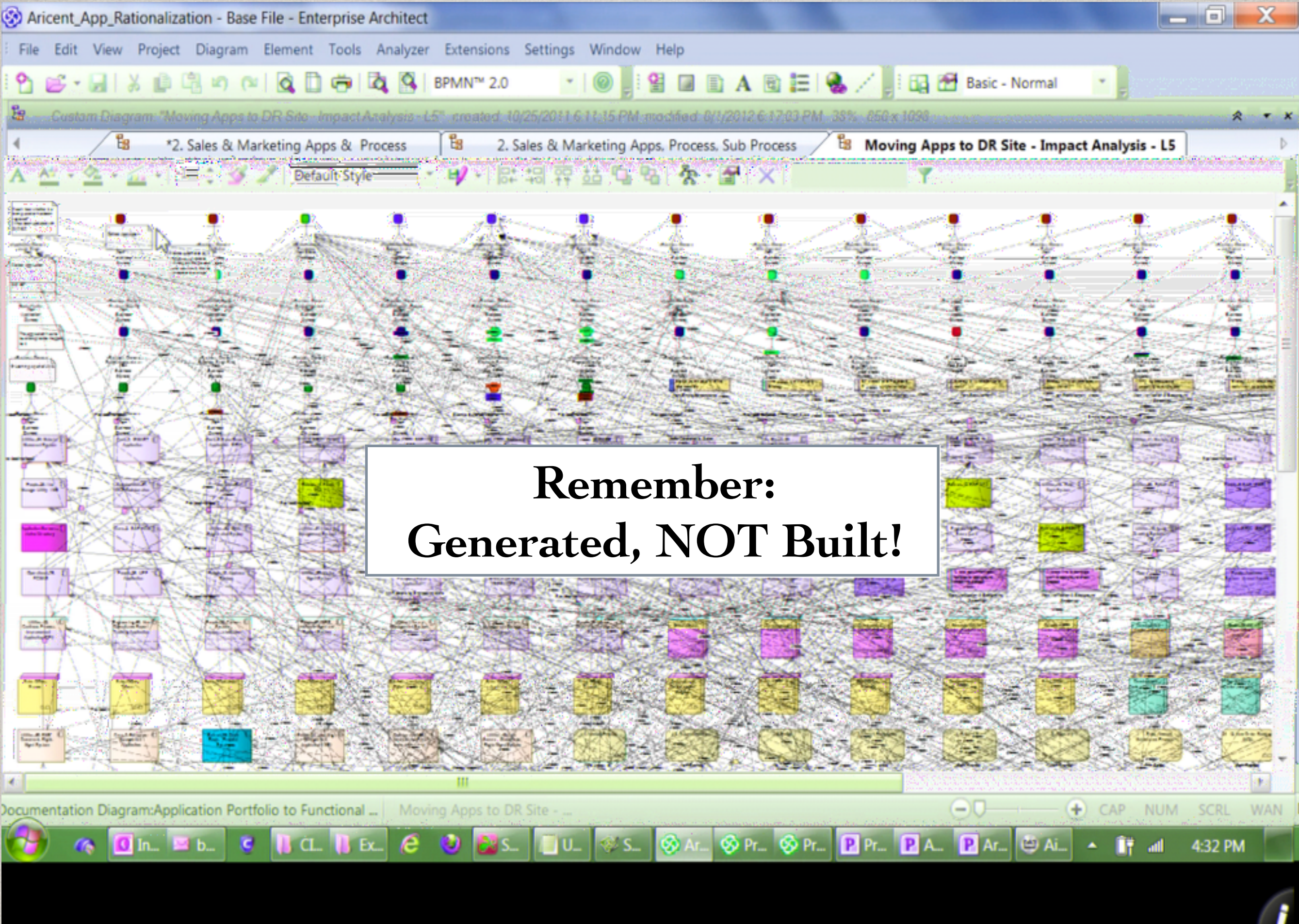
Technical
Personnel
(Business
Implementation)



Enterprise
Personnel
(Enterprise
Architecture)







Remember:
Generated, NOT Built!

THE KEY

1. Single-variable, precisely unique, relevant (not arbitrary),
ontologically-based components.
2. Binary Relationships (only two components at a time).

Exercise - Deconstruct Business Process Model for multiple Targets

MISSING in this case

Enterprise architecture Plan

1. Which process needs to be automated?
- Target driven by change in "Activity" or process flows

2. What roles need to be added or removed for improving the outcome?
- Target driven by change in "Rules"

3. Change "process" flows keep others constant
- Which process needs to be automated?

4. Change the "responsibilities" keep others constant
- What roles need to be added or removed for improving the outcome?

5. Change the "distribution" keep others constant
- Can we minimize the process flow across the geographical locations for cost & time savings?

6. Change the "rules" keep others constant
- Which rule is mandatory for workflow to be reliable?

7. Optimize the "events" keep others constant
- Which events will improve the timeliness of the process?

8. Change the "data" elements keep others constant
- Is the input (data) likely to change?
- Are all the activities, rules, roles, data, location, events are connected?

Only two interrogatives being used instead of six

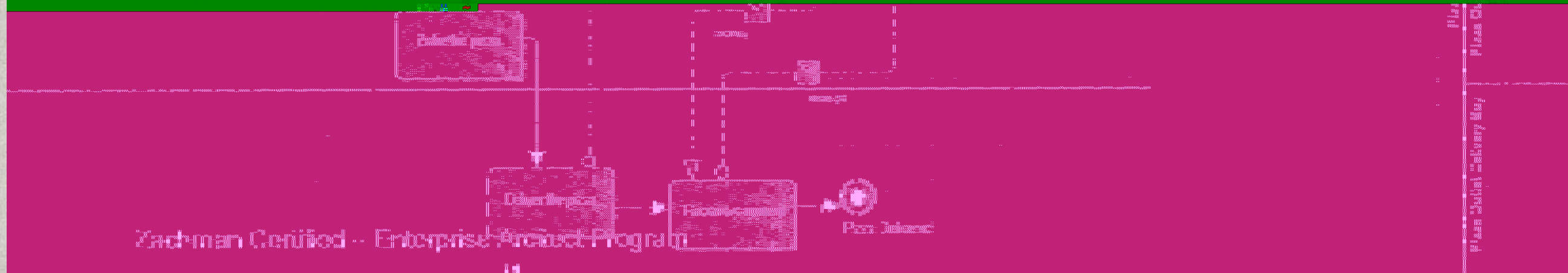
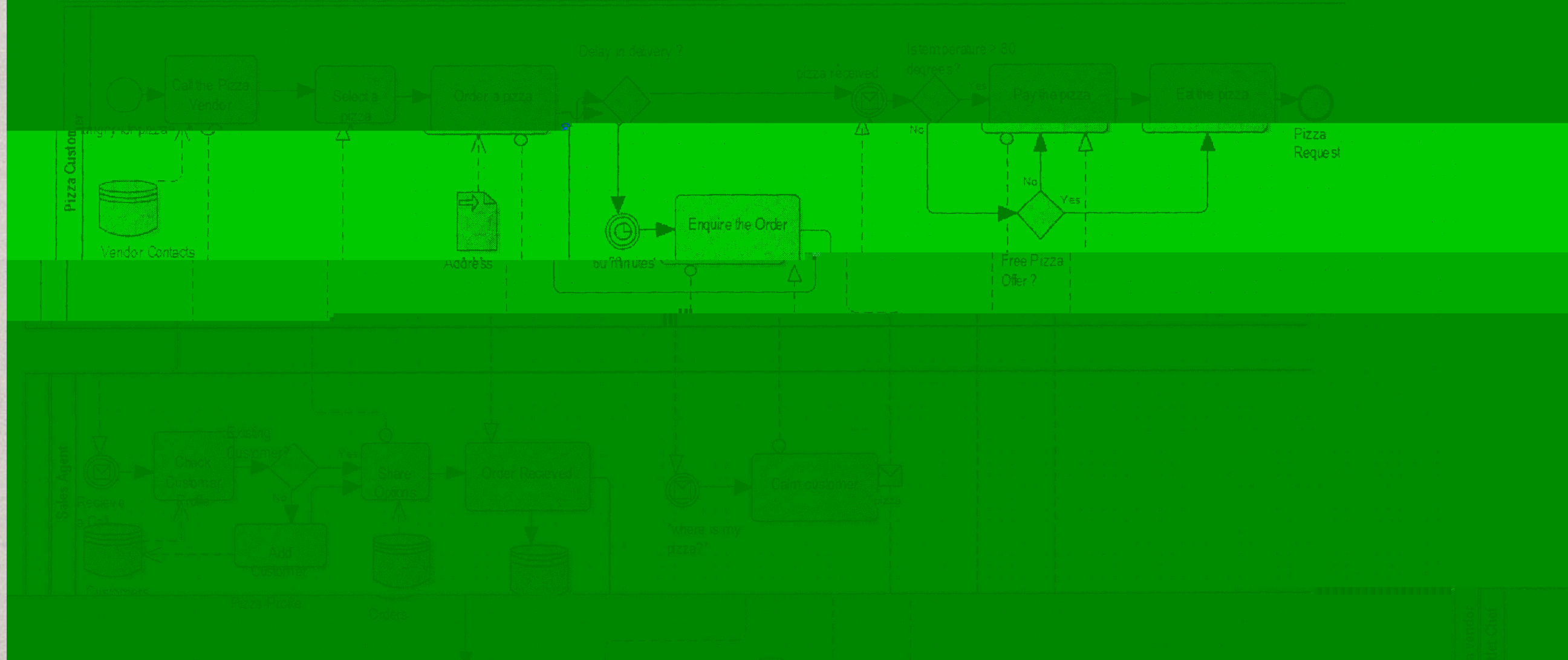
SIX interrogatives being used. Hundreds of permutations & combinations

Copyright © 2012 iCMG

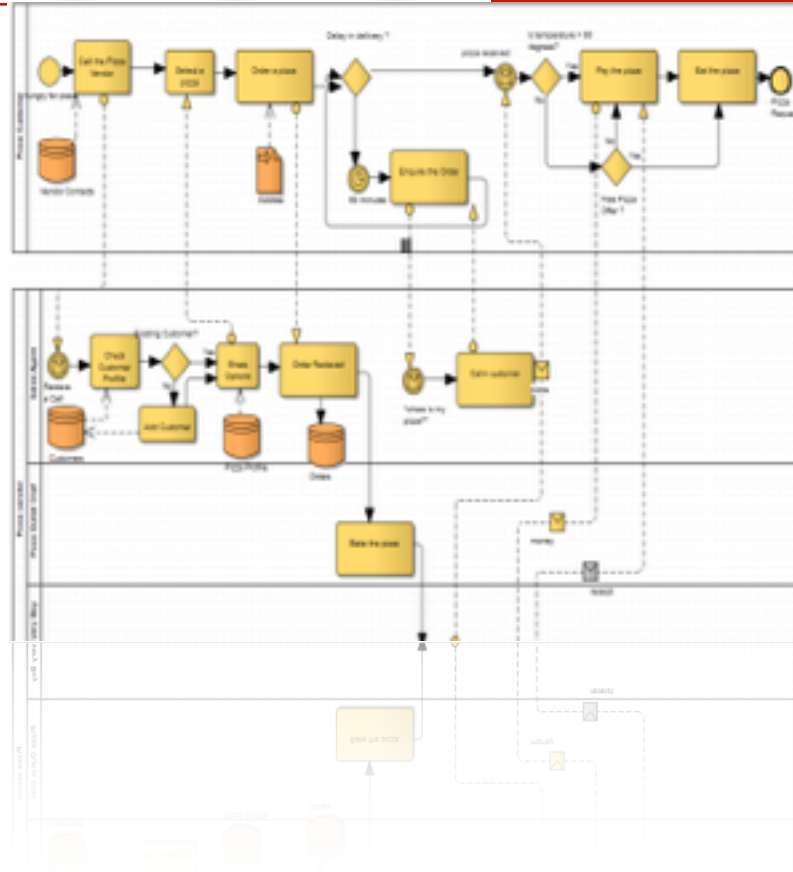
Nömbigg, may be? Not sure?

Exercise – Pizza Delivery Process

Use the Enterprise Architect Framework: 5 Data Elements, 5 Roles, 4 Rules, 3 Locations

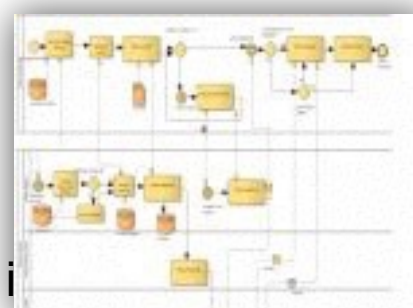
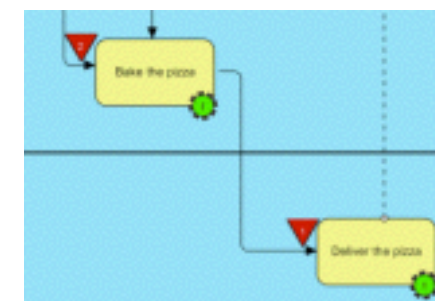
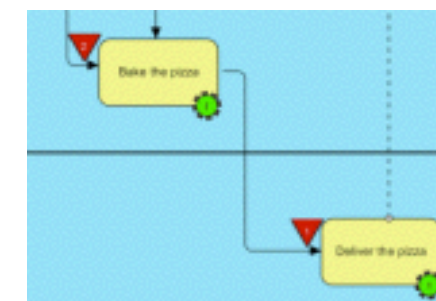
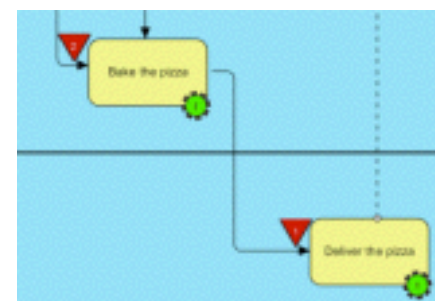
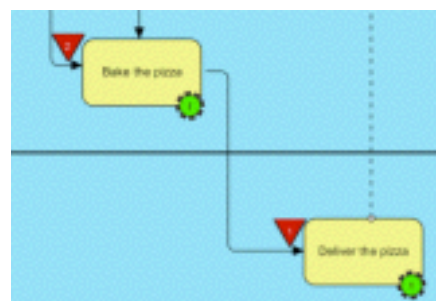
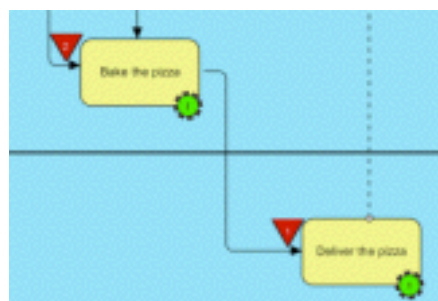
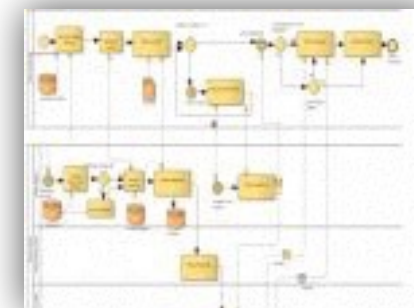


Exercise 3



DATA What (Things)	FUNCTION How (Process)	NO. THINGS Where (Location)	PEOPLE Who (People)	TIME When (Time)	MOTIVATION Why (Motivation)
Customer	Add Customer	i-Pizza Customer Location	i-Pizza Customer	Pizzeria in i-Cat	Order a pizza
Vendor Contacts	Make the price			Hungry for pizza	
Address	Call the Pizza Service	i-Pizza Service	i-Pizza Service	30-minute opt.	Order a pizza
Orders	Order a pizza			price-related	
Price Profile	Pay the price	i-Pizza Service	i-Pizza Service	When is my pizza arriving?	Interpretation
	Deliver the pizza			Pizza Received	
	Receive payment			Pizza Delivered	

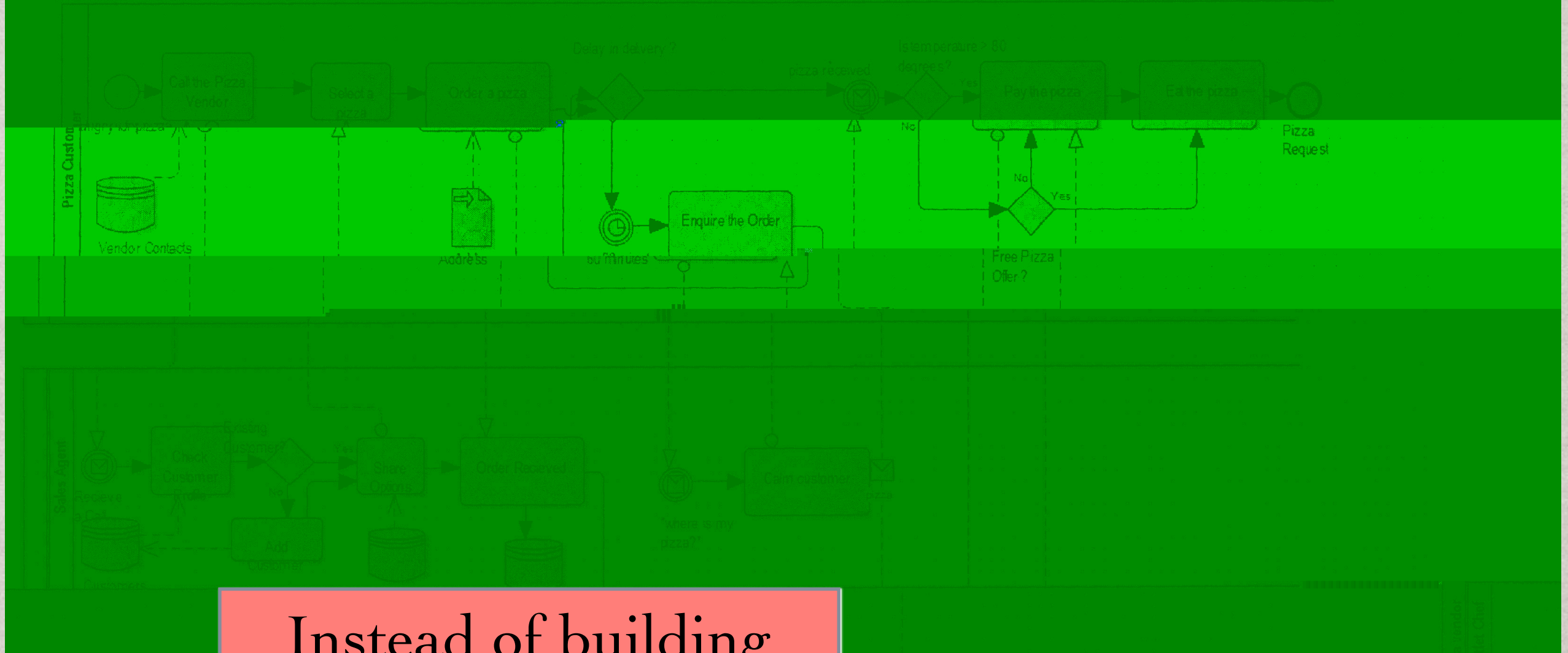
Exercise 3



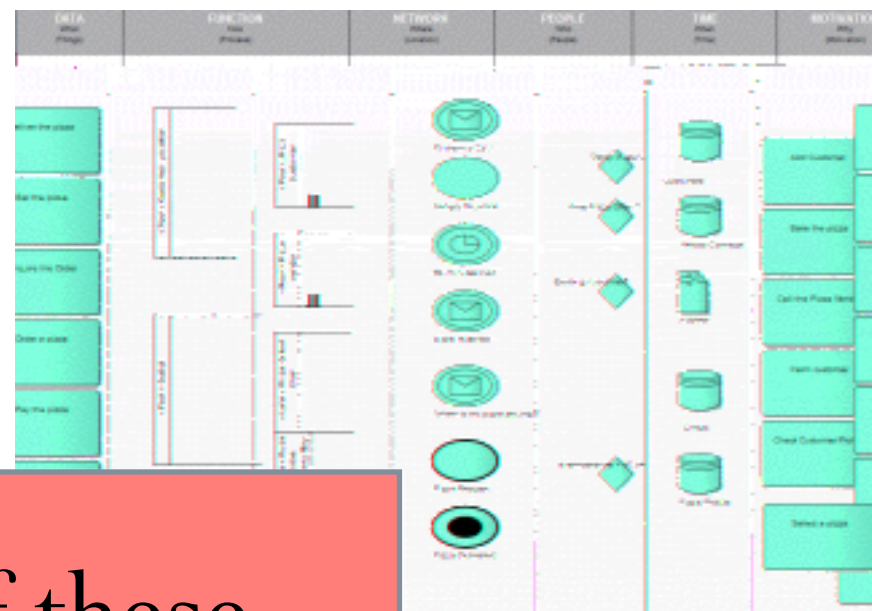


Exercise – Pizza Delivery Process

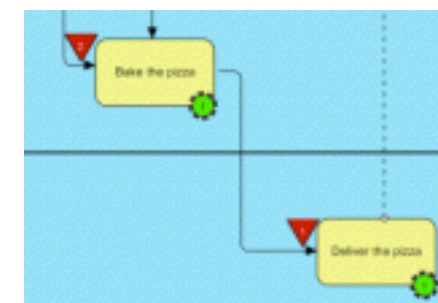
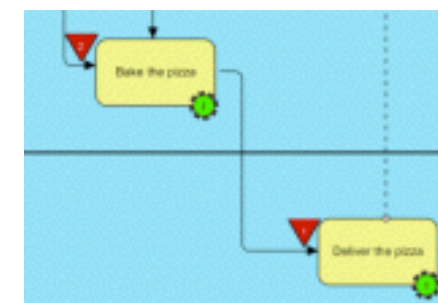
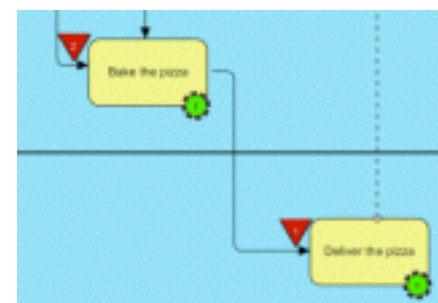
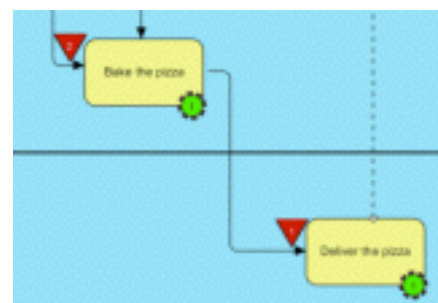
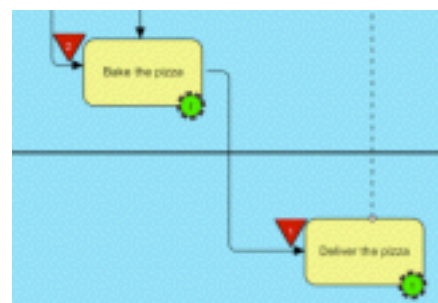
This Process consists of 15 Activities, 5 Data Elements, 5 Roles, 4 Rules, 3 Locations



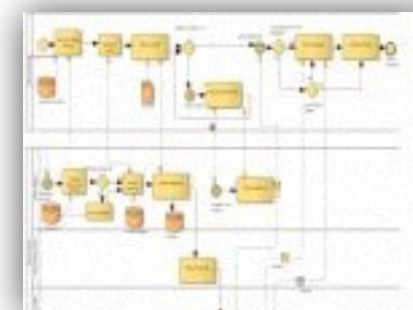
Instead of building
ONE of these



Build “n” of these.



Pick the one you like.



THE KEY

1. Single-variable, precisely unique, relevant (not arbitrary),
ontologically-based components.
2. Binary Relationships (only two components at a time).

Remember! This is a PRIMITIVE (single-variable) Model used for Engineering.

It cannot be used for implementations which require COMPOSITE (multi-variable) Models.

(Some possible COMPOSITE integration relationships may be shown at the periphery of the model. The COMPOSITE implementation "view" would be created by re-using components of other Enterprise-wide, "engineered" PRIMITIVES.

What

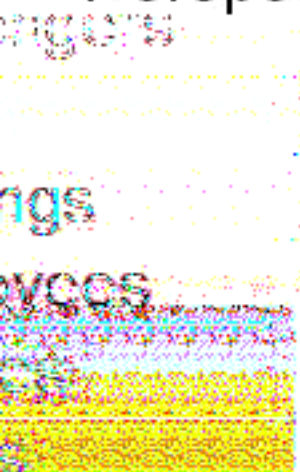
(Column 1)

Inventory Identification

Row 1
Executive
Perspective

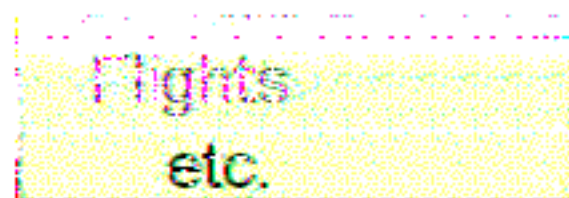
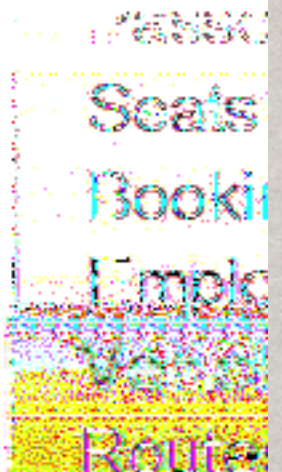
Airplanes
Airplane Types
Airports
Gates
Passengers

Scope
Contexts



Composites
There can be composite
relationships with any or
all other Row 1 Cells and
each one can have one
or more instances in Row 2.

Note:
A Transportation Case
overlaid (Drives)
Collection Things (Airline)
(Likely have serial numbers)
A List - Scope



Row 2 Instances AS IS may or may not have anything
to do with Owners, Designers, Users, etc. etc.
and these are not necessarily related to the
Inventory

Note: This sample model is
meant to illustrate the form of
the examples. It is not
necessarily the correct

Level of Detail - High
Not a real (or fake) model
as a model has to be
No Recurring Concepts

Sets

ENTERPRISE FRAMEWORK

Remember! This is a PRIMITIVE (single-variable) model used for Engineering.

It cannot be used for implementations which require COMPOSITE (multi-variable) Models.

(Some possible COMPOSITE integration relationships may be shown at the periphery of the model. The COMPOSITE information "layer" would be shared by re-using components of other Enterprise-wide Architectural PRIMITIVES.

How

(Column 2)

Process Identification

Row 1
Executive
Perspective

Note:

Air Transportation Case
Processes (Transformations)
(Transitive Verb-Object)

A List - Scope

Level of Detail = High

Abstract (no instances)

As simple as possible

No Recurring Concepts

Acquire Routes
Schedule Flights
Sell Bookings
Reserve Seats
Train Employees
Fly Airplanes
Schedule Crews
Repair Facilities
Develop Markets
Maintain Airplanes
etc.

Scope
Contexts

Composites

There can be composite
relationships with any or
all other Row 1 Cells and
with the Cell below and
Instances in Row 6.

Row 6 Instances AS IS may or may not have anything
to do with Owner's, Designer's, Builders perceptions
until these are made explicit and transformed to Row 6

Note: This sample model is
meant to illustrate the form of
the expected Primitive, not
necessarily the content.

Process

Flows

Inventory Sets

Airplanes
Airplane Types
Airports
Gates
Passengers
Shareholders
Local Carriers
Seats
Bookings
Routes
Employees
Vehicles
Flights
etc.

Process Flows

Acquire Routes
Schedule Flights
Reserve Seats
Train Employees
Fly Airplanes
Schedule Crews
Repair Facilities
Develop Markets
Maint. Airplanes
Load Airplanes
Release Flights
Develop Flt. Plans
Schedule Maint.
etc.

Distr. Networks

Airplane Network
Parts Distr. Net.
Communications
Freight Net.
Airport Network
(Runways, etc.)
Regulatory Net.
Passenger Net.
Personnel Net.
Catering Net.
etc.

Respon Assmts

Timing Cycles

Motive Intent.

Pilots	Flight Cycle	Equip. Utilization
Co-pilots	Customer Cycle	New Markets
Engineers	Maintenance Cyc.	Revenue Growth
Flt. Attend.	Telephone Wait C.	Exp. Reduction
Reservations	Airplane Turnaround	Cust Convenience
Aircraft Maint.	De-Icing Cycle	Cust. Satisfaction
Flight Scheduling	Air Traffic Cntl. C.	Labor Contracts
Airport Ops.Mgt	Tarmac Cycle	Regulatory Comp
Customer Service	Airplane Cycle	New Capital
Marketing	Baggage Handling C.	Load Factor
Sales	Security (TSA) Cycle	Route Optimize
Flight Dispatch	Planning Cycle	Flight Expansion
Accounting	Budget Cycle	Acquisition
etc.	etc.	etc.

THE KEY

1. Single-variable, precisely unique, relevant (not arbitrary),
ontologically-based components.
2. Binary Relationships (only two components at a time).

INTRODUCTION TO
ENTERPRISE ARCHITECTURE

METHODOLOGY
FOR SOLVING
GENERAL MANAGEMENT
PROBLEMS

JOHN A. ZACHMAN
ZACHMAN INTERNATIONAL

THE PROCESS

1. Select General Management Problem.
2. Factor out Primitive (**Single-variable**) Components, sort into Zachman Framework Cells (make Lists).
3. Define “binary” (only two at a time) dependencies (horizontal and vertical) between Primitive (**Single-variable**) Components.
4. Create Composite (**Multi-variable**) “snapshot” of problem area and diagnose.
5. Pose new Composite (**Multi-variable**) scenarios (change Lists and/or change dependencies) and re-compose multiple targets.
6. Add time and cost to Primitive (**Single-variable**) Components and simulate alternatives, perform risk analysis, identify resource rqmts, etc.

(CEO picks solution, assigns responsibilities for “quick-fix”, identifies subsequent CEO problem. Then re-iterate Steps 1 - 6.)

THE PROCESS (CONT.)

(Architecture process proceeds in parallel:)

7. Pick several Cells in different Columns in some Row and assign modeling experts to build out complete (thing-relationship-thing) Primitive (**Single-variable**) Models, verifying horizontal alignment.
8. Create complete Composite (**Multi-variable**) integration for Row ensuring horizontal alignment. (Does each Cell have all components required for reuse in adjoining Cells for creating Composites.)
9. Have Columnar modeling experts transform Primitive (**Single-variable**) Models to next Cell below, ensuring vertical “alignment” and iterate until all Cells in the Column are transformed and vertically aligned.
10. Transform Row 5 Primitives (**Single-variables**) to Row 6 implementations (**Multi-variables**) using either machines (automated) or people (manual).
11. Add Primitive Components by Cell from next problem and reiterate Steps 7 - 11.

THE PROCESS (CONT.)

12. Institutionalize this process and govern Architecture as follows:
 - a. prohibit redundancy except where explicitly controlled.
 - b. maintain horizontal and vertical alignment
 - c. use Primitive (**Single-variable**) Model inventory as base for managing ENTERPRISE changes.
 - d. ensure **EVERY** new implementation Composite reuses components of Primitive models and migrate legacy to Architected Enterprise. (See Workshop “Migration Strategy”.)
13. Acquire subject matter expertise for building additional Primitive (**Single-variable**) Models to be added to the Enterprise Architecture capability inventory.

Key: Single-Variable, PRIMITIVE Models, and Binary Relationships

For details see Level 2 Zachman Certification at www.Zachman.com

Note: This is the same process, somewhat abbreviated, and executed by students in the 4 day Zachman Level 1 Certification Workshop.

E N T E R P R I S E A R C H I T E C T U R E

CONCLUSIONS

J O H N A . Z A C H M A N
Z A C H M A N I N T E R N A T I O N A L

RESEARCH LESSONS

- A. It is possible to solve General Management problems very quickly with a small subset of Primitive components (simply Lists and their inter-dependencies short of the complete Primitive Models)
- B. Different complex, composite constructs can be created dynamically, virtually cost-free, from the inventory of Primitive Lists for addressing subsequent General Management problems.
- C. Many scenarios can be evaluated to test strategy alternatives before making commitments.

PROFOUND SIGNIFICANCE

- A. It alters the concept of Enterprise Architecture from one of building models to one of solving General Management problems.
- B. Proves the validity of the Primitive Model concept: from a finite inventory of Primitive Concepts you can dynamically create a virtually infinite number of Enterprise implementation Composites.
- C. Buys the time for “the experts” to build out the complete Enterprise Architecture (Thing-Relationship-Thing) Primitive Models iteratively and incrementally.
- D. Builds significant credibility for the Information Technology community.
- E. Establishes the basis for an Enterprise Architecture Profession.

CHALLENGE TO ENTERPRISE ARCHITECTS

Reframe the concept of Enterprise Architecture ...

It is not about building models!

It is about solving Enterprise problems
while iteratively and incrementally building
out the inventory of complete, reusable,
Primitive Models that constitute:

Enterprise Architecture.

JAY W. FORRESTER

"Although social systems are more complex than physical systems, they belong to the same class of high-order, non-linear, feedback systems as do physical systems.

People do not accept the idea that families, corporations, and governments belong to the same class of dynamic structures as do chemical refineries and autopilots for aircraft.

"Organizations built by committee and intuition perform no better than would an airplane built by the same methods ... As in a bad airplane design, which no pilot can fly successfully, such badly designed corporations lie beyond the ability of real-life managers.

"I anticipate future management schools devoted to 'enterprise design'. ... A fundamental difference exists between an enterprise operator and an enterprise designer. A manager runs an organization, just as a pilot runs an airplane. Success of a pilot depends on an aircraft designer who created a successful airplane. ...who designed the corporation that a manager runs?"

"Designing the Future" by Jay W. Forrester 12/15/98

1965 SYSTEMS PROBLEMS

1. Didn't meet Requirements. (not "aligned")
2. The data was no good:
 - Not consistent from system to system.
 - Not accurate.
 - Not accessible.
 - Too late.
3. Couldn't change the system. (Inflexible)
4. Couldn't change the technology. (Not adaptable)
5. Couldn't change the business. (Couldn't change the system or the technology so couldn't change business.)
6. Little new development (80% \$ for maintenance)
7. Took too long.
8. Cost too much.
9. Always over budget.
10. Always missed schedules.
11. DP budget out of control.
12. Too complicated - can't understand it, can't manage it.
13. Just frustrating.

(Adapted from Doug Erickson)

2015 SYSTEMS PROBLEMS

1. Didn't meet Requirements. (not "aligned")
2. The data was no good:
 - Not consistent from system to system.
 - Not accurate.
 - Not accessible.
 - Too late.
3. Couldn't change the system. (Inflexible)
4. Couldn't change the technology. (Not adaptable)
5. Couldn't change the business. (Couldn't change the system or the technology so couldn't change business.)
6. Little new development (80% \$ for maintenance)
7. Took too long.
8. Cost too much.
9. Always over budget.
10. Always missed schedules.
11. IT budget out of control.
12. Too complicated - can't understand it, can't manage it.
13. Just frustrating.

(Adapted from Doug Erickson)

IT'S FUNNY...

COBOL didn't fix those problems!

MVS didn't fix those problems!

Virtual Memory didn't fix those problems!

IMS, DB2, Oracle, Sybase, Access, Fortran, PL/1, ADA, C++, Visual Basic, JAVA 2, 360's, 390's, MPP's, DEC VAX's, H200's, Crays, PC's, MAC's, Distributed Processing, didn't fix those problems!

Word, Excel, Powerpoint, Outlook Express, eMAIL, DOS, Windows 95, 98, 2000, NT, ME, XP, Unix, Linux, Object Oriented, COM, DCOM, CORBA, EDI, HTML, XML, UML, the Internet, B2B, B2C, Portals, Browsers didn't fix those problems!

IEF, IEW, ADW, ERWIN, POPKIN, Rational, Casewise, Rochade, Platinum, Design Bank, Data Warehouse, SAP, Baan, Peoplesoft, Oracle Financials, BSP, ISP, EAP, EAI didn't fix those problems!

And, I doubt that Web Services, .Net, Agile Programming, Service Oriented Architecture, Cloud Computing, BigData or I.B.Watson (whoever that is) is going to fix the problems.

IT MAKES ONE WONDER IF THERE ACTUALLY IS A TECHNICAL SOLUTION TO THE PROBLEMS!!!

ENGINEERING PROBLEM

I'm not saying that there is anything wrong with any of these technologies.

In fact, any or all of them may well be very good ...

In fact, you may not be able to solve the Enterprise problem without employing some of these technologies.

However, The Enterprise problem is an ENGINEERING problem, NOT a technical problem.

My perception is that it is going to take actual work, ENGINEERING work, to solve the problems. My plan would be to start building out an inventory of models, PRIMITIVE MODELS, iteratively and incrementally, engineering them for alignment, integration, flexibility, reduced time-to-market, etc., etc.

What would be YOUR plan for solving the problems???